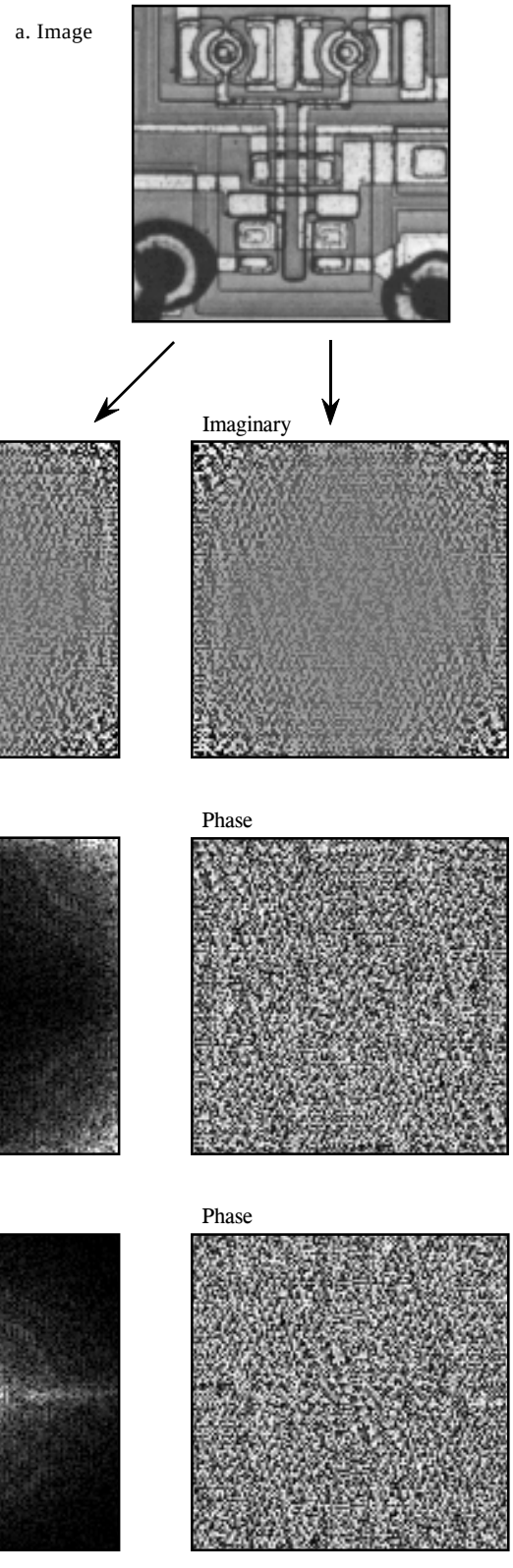
FIGURE 24-9
Frequency spectrum of an image. The example image, shown in (a), is a microscopic photograph of the silicon surface of an integrated circuit. The frequency spectrum can be displayed as the real and imaginary parts, shown in (b), or as the magnitude and phase, shown in (c). Figures (b) & (c) are displayed with the low-frequencies at the corners and the high-frequencies at the center. Since the frequency domain is periodic, the display can be rearranged to reverse these positions. This is shown in (d), where the magnitude and phase are displayed with the low-frequencies located at the center and the high-frequencies at the corners.



b. Frequency spectrum displayed in rectangular form (as the real and imaginary parts).

c. Frequency spectrum displayed in polar form (as the magnitude and phase).

d. Frequency spectrum displayed in polar form, with the spectrum shifted to place zero frequency at the center.